Aligning Small LLMs with GRPO for Strict JSON Generation

Rule-Based Reward Optimization for Schema-Conformant Output

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Problem & Objective

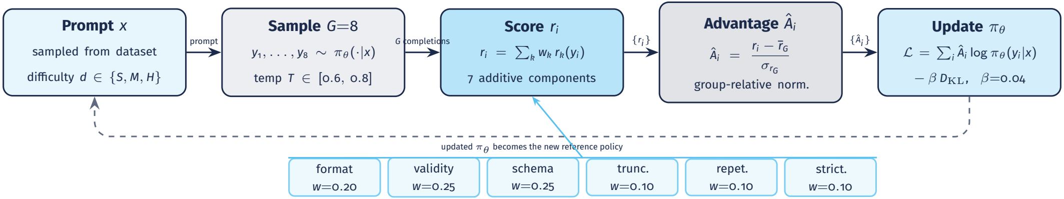
- Small LLMs ($<$2B parameters) frequently fail to produce valid, schema-conformant JSON
- Common errors: unclosed brackets, type violations, conversational filler
- **Goal:** use GRPO with rule-based rewards to reach $>$90% Pass@1 on strict JSON
- No neural reward model $\rightarrow$ dense, deterministic, interpretable signal

Models & Hardware Configuration

Model	Params	Architecture
SmolLM2-135M-Instruct	135M	LLaMA-like
SmolLM2-360M-Instruct	360M	LLaMA-like
Qwen2.5-0.5B-Instruct	0.5B	Qwen2.5
TinyLlama-1.1B-Chat	1.1B	LLaMA 2
Gemma-2-2B-it	2B	Gemma 2

- 4-bit NF4 quantization
- LoRA $r=16$, $\alpha=32$
- $1\times$ NVIDIA L40S
- 2 500 training steps / model

Group Relative Policy Optimization (GRPO)



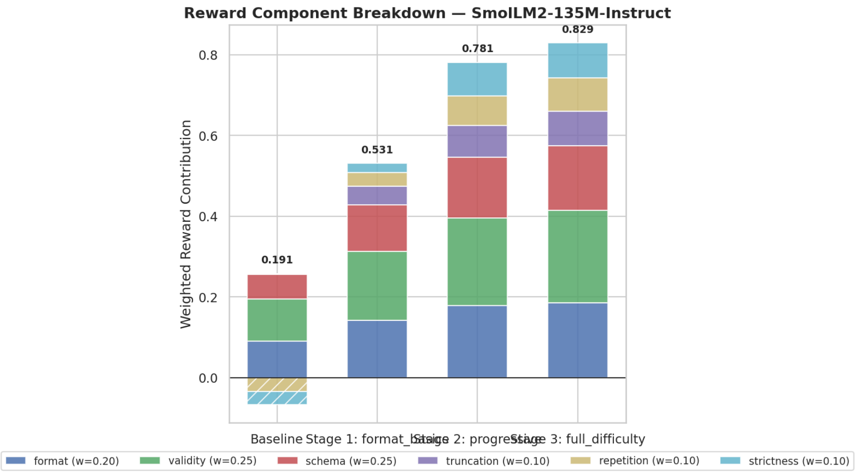
- No neural reward model — scoring is deterministic and interpretable
- Additive composition prevents zero-advantage collapse: gradient is always non-zero
- KL penalty ($\beta{=}0.04$) keeps π_θ close to the reference policy

Rule-Based Reward System

Component	Range	Think w	NoThink w	What it measures
`format`	$[0, 1]$	0.10	0.20	“‘`json`“‘ fence: 1.0 $\mid$ generic fence: 0.5 $\mid$ none: 0.0
`validity`	$[0, 1]$	0.15	0.25	JSON parseability; partial credit by error position (0.20–0.70 if invalid)
`schema`	$[0, 1]$	0.20	0.25	Keys, types, nesting depth, required fields; 1.0 if valid but unconstrained
`reasoning`	$[-0.5, 1]$	0.25	0.0	<think> block quality: penalises missing/copied placeholder
`truncation`	$[-1, 1]$	0.10	0.10	Unclosed braces / unterminated strings / trailing commas
`repetition`	$[-1, 1]$	0.10	0.10	Token loops, repeated lines, duplicate blocks
`strictness`	$[-1, 1]$	0.10	0.10	Extra text outside JSON $>$50% of output

Additive composition — no hard gating, gradient is always non-zero.

Reward Breakdown — SmolLM2-135M



No-Think / Curriculum — component contributions at final checkpoint

Training Strategies

	Standard	Curriculum
No-Think	✓	✓
Think	✓	✓

Think: model generates <think> reasoning before JSON

Curriculum: 3 progressive stages

Stage	Steps	S / M / H	Temp
1 – Format Basics	800	35 / 55 / 10%	0.8
2 – Progressive	800	15 / 35 / 50%	0.7
3 – Full Difficulty	900	10 / 25 / 65%	0.6

S = Simple M = Medium H = Hard

Parametric Synthetic Dataset

- 24 parameterized templates (8 per difficulty tier)
- 1 500 samples/stage × 3 stages = **4 500** training samples
- Eval: **300 samples** — 100 simple / 100 medium / 100 hard
- **No ground-truth JSON**
  reward functions evaluate directly

Difficulty tiers

Simple — flat key-value objects, typed arrays

Medium — nested objects, config files, form schemas

Hard — JSON Schema draft-07, paginated APIs, OpenAPI specs

Table 1 — Baseline Pass@1 (No-Think)

Model	Overall	Simple	Medium	Hard
SmolLM2-135M	39.00%	58.00%	39.00%	20.00%
SmolLM2-360M	77.67%	85.00%	86.00%	62.00%
Qwen2.5-0.5B	92.67%	95.00%	95.00%	88.00%
TinyLlama-1.1B	79.33%	89.00%	76.00%	73.00%
Gemma-2-2B	**98.33%**	**100%**	**100%**	**95.00%**

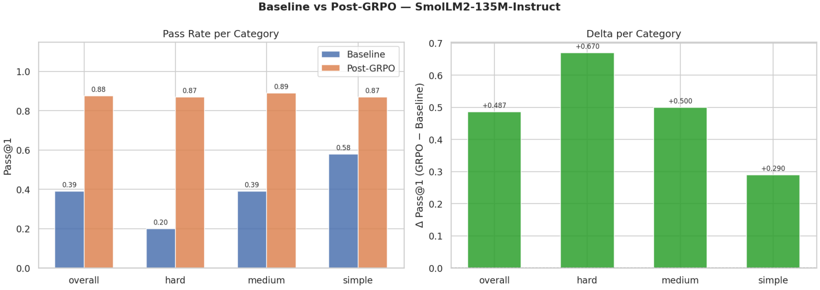
300-sample balanced eval set (100 simple / 100 medium / 100 hard)

Table 2 — Post-Training Pass@1 (No-Think / Standard, 2 500 steps)

Model	Overall	Simple	Medium	Hard	Δ
SmolLM2-135M	87.67%	87.00%	89.00%	87.00%	+48.67
SmolLM2-360M	96.67%	97.00%	97.00%	96.00%	+19.00
Qwen2.5-0.5B	97.33%	99.00%	95.00%	**98.00%**	+4.67
TinyLlama-1.1B	**97.67%**	**100%**	96.00%	97.00%	+18.33
Gemma-2-2B	96.67%	**100%**	**99.00%**	91.00%	−1.67

Gemma-2-2B regresses (−1.67 pp) — mixed-difficulty training without curriculum

SmolLM2-135M — Baseline vs. GRPO (No-Think / Standard)



Pass@1 by difficulty tier before and after 2 500 GRPO steps

Table 3 — Baseline Pass@1 (No-Think / Curriculum)

Model	Overall	Simple	Medium	Hard
SmolLM2-135M	33.00%	49.00%	32.00%	18.00%
SmolLM2-360M	73.67%	82.00%	84.00%	55.00%
Qwen2.5-0.5B	89.33%	**98.00%**	91.00%	79.00%
TinyLlama-1.1B	79.33%	89.00%	83.00%	66.00%
Gemma-2-2B	**97.00%**	**100%**	**98.00%**	**93.00%**

Slight variance vs. Table 1 due to different generation seed

Table 4 — Post-Training Pass@1 (No-Think / Curriculum, 2 500 steps)

Model	Overall	Simple	Medium	Hard	Δ
SmolLM2-135M	89.00%	90.00%	86.00%	91.00%	+56.00
SmolLM2-360M	95.33%	93.00%	95.00%	**98.00%**	+21.67
Qwen2.5-0.5B	97.33%	**100%**	95.00%	97.00%	+8.00
TinyLlama-1.1B	**99.33%**	**100%**	**99.00%**	**99.00%**	+20.00
Gemma-2-2B	97.67%	**100%**	**99.00%**	94.00%	+0.67

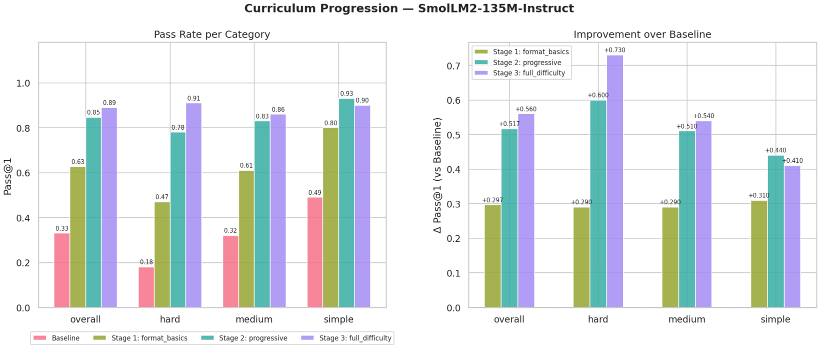
Curriculum protects Gemma-2-2B from regression (+0.67 vs. −1.67 in standard)

Table 5 — Curriculum Stage Progression (No-Think)

Model	Baseline	Stage 1	Stage 2	Stage 3
SmolLM2-135M	33.00%	62.67%	84.67%	**89.00%**
SmolLM2-360M	73.67%	79.67%	91.33%	**95.33%**
Qwen2.5-0.5B	89.33%	95.67%	97.00%	**97.33%**
TinyLlama-1.1B	79.33%	94.33%	98.33%	**99.33%**
Gemma-2-2B	**97.00%**	**98.33%**	97.67%	97.67%

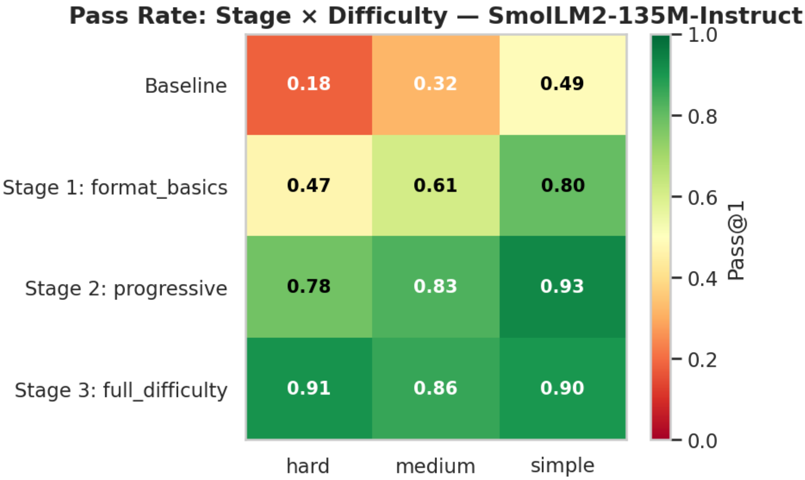
Monotonic improvement for all models; Gemma peaks at Stage 1

SmolLM2-135M — Curriculum Progression (No-Think)



33% → 62.7% → 84.7% → 89.0% across 3 curriculum stages

SmolLM2-135M — Difficulty Heatmap (No-Think / Curriculum)



Pass@1 per difficulty tier at each curriculum stage

Table 6 — Baseline Pass@1 (Think)

Model	Overall	Simple	Medium	Hard
SmolLM2-135M	31.33%	44.00%	33.00%	17.00%
SmolLM2-360M	69.33%	79.00%	80.00%	49.00%
Qwen2.5-0.5B	90.33%	**98.00%**	92.00%	81.00%
TinyLlama-1.1B	85.67%	90.00%	87.00%	80.00%
Gemma-2-2B	**96.67%**	99.00%	**98.00%**	**93.00%**

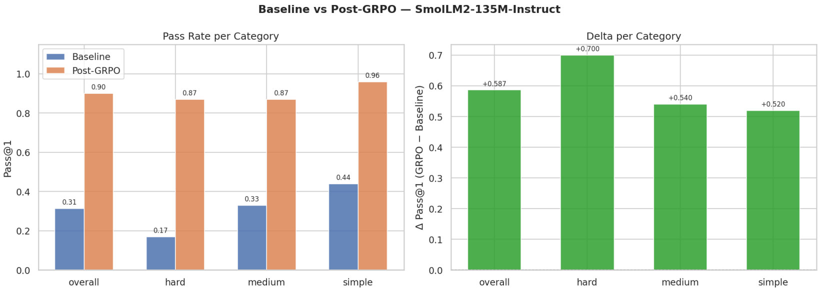
<think> prompt lowers baseline for small models (360M: 77.67% $\rightarrow$ 69.33%)

Table 7 — Post-Training Pass@1 (Think / Standard, 2 500 steps)

Model	**Overall**	**Simple**	**Medium**	**Hard**	**Δ**
SmolLM2-135M	90.00%	96.00%	87.00%	87.00%	+58.67
SmolLM2-360M	95.67%	96.00%	97.00%	94.00%	+26.33
Qwen2.5-0.5B	97.67%	**100%**	**98.00%**	95.00%	+7.33
TinyLlama-1.1B	**98.67%**	99.00%	**98.00%**	**99.00%**	+13.00
Gemma-2-2B	96.00%	99.00%	**99.00%**	90.00%	−0.67

SmolLM2-135M: largest absolute gain in the study (+58.67 pp)

SmolLM2-135M — Baseline vs. GRPO (Think / Standard)



Think / Standard — reasoning tokens unlock the highest peak for 135M

Table 8 — Baseline Pass@1 (Think / Curriculum)

Model	Overall	Simple	Medium	Hard
SmolLM2-135M	31.67%	51.00%	25.00%	19.00%
SmolLM2-360M	71.33%	76.00%	79.00%	59.00%
Qwen2.5-0.5B	91.67%	**99.00%**	89.00%	87.00%
TinyLlama-1.1B	83.67%	96.00%	84.00%	71.00%
Gemma-2-2B	**96.33%**	**100%**	**96.00%**	**93.00%**

Same capability hierarchy as Table 6; slight sampling variance

Table 9 — Post-Training Pass@1 (Think / Curriculum, 2 500 steps)

Model	**Overall**	**Simple**	**Medium**	**Hard**	**Δ**
SmolLM2-135M	89.33%	94.00%	88.00%	86.00%	+57.67
SmolLM2-360M	93.00%	92.00%	97.00%	90.00%	+21.67
Qwen2.5-0.5B	**98.00%**	**100%**	**97.00%**	**97.00%**	+6.33
TinyLlama-1.1B	97.67%	**100%**	96.00%	**97.00%**	+14.00
Gemma-2-2B	97.67%	99.00%	**99.00%**	95.00%	+1.33

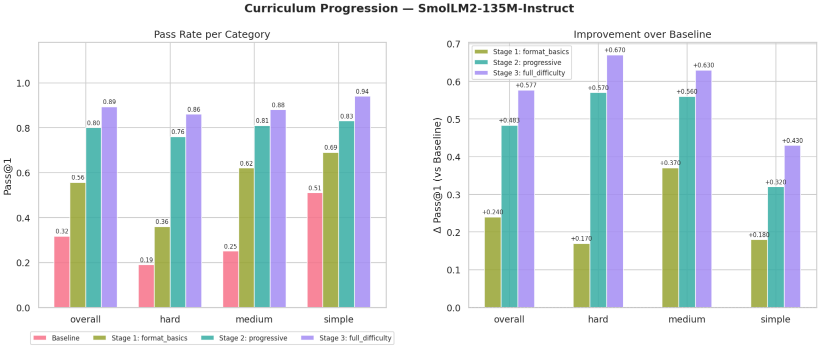
Curriculum fixes Gemma-2-2B: +1.33 pp vs. −0.67 in standard

Table 10 — Curriculum Stage Progression (Think)

Model	Baseline	Stage 1	Stage 2	Stage 3
SmolLM2-135M	31.67%	55.67%	80.00%	**89.33%**
SmolLM2-360M	71.33%	78.33%	90.33%	**93.00%**
Qwen2.5-0.5B	91.67%	94.67%	93.33%	**98.00%**
TinyLlama-1.1B	83.67%	94.33%	**97.00%**	97.67%
Gemma-2-2B	**96.33%**	83.00%	66.00%	97.67%

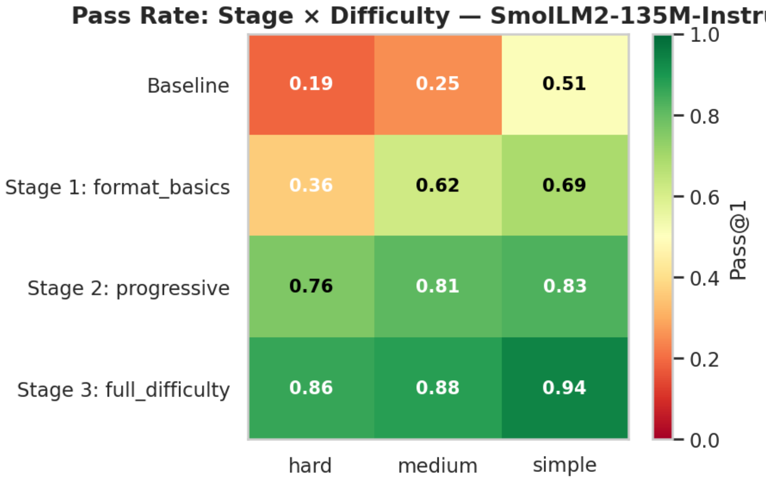
Gemma-2-2B U-curve: 96% → 83% → 66% → 98% — temporary collapse

SmolLM2-135M — Curriculum Progression (Think)



31.7% → 55.7% → 80.0% → 89.3% across 3 curriculum stages

SmolLM2-135M — Difficulty Heatmap (Think / Curriculum)



Pass@1 per difficulty tier at each curriculum stage

Lessons & Limitations

Technical lessons

- **Additive rewards $>$ gating**
  zero-advantage collapse if all completions tie $\rightarrow$ gradient vanishes
- **Graduated partial credit** matters
  binary validity $\rightarrow$ no signal for near-valid JSON
- **4-bit NF4 + LoRA** ($r=16$) is sufficient
  $<$2% of parameters trained, single GPU
- **Weight redistribution** when Think is off
  preserves component ratios automatically

Limitations

- JSON only — not directly generalizable to other formats
- Fully synthetic dataset
  $\rightarrow$ potential real-world gap
- 2 500 steps / single GPU
  longer runs may yield more

Conclusions

- GRPO + rule-based rewards brings all 5 models to 87–99% Pass@1 — closing the gap to much larger models
- **Standard training** is optimal for weak models:
  SmolLM2-135M +48.67 pp, TinyLlama-1.1B +18.33 pp
- **Curriculum** is a safety net for strong models:
  Gemma-2-2B flips from −1.67 pp (standard) to +0.67 pp
- **Think mode** yields $<$2 pp marginal gain — extra latency not justified for sub-1B models
- All configurations benefit from **identical hyperparameters**, confirming that reward design matters more than tuning